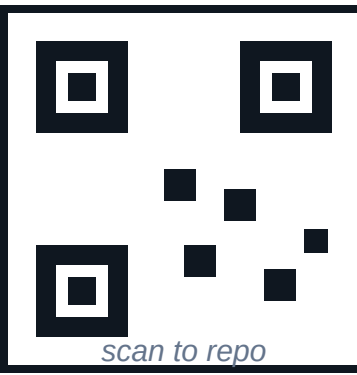


Statistical Screening for Poisoned Machine-Learning Training Data



Detection and Quarantine of Suspicious Samples at the Data-Ingestion Boundary

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Security Engineering • AI / Agent Security • Cloud Security

MIT • Python 3.12 • HEAD b45d662 • verified 2026-09-26

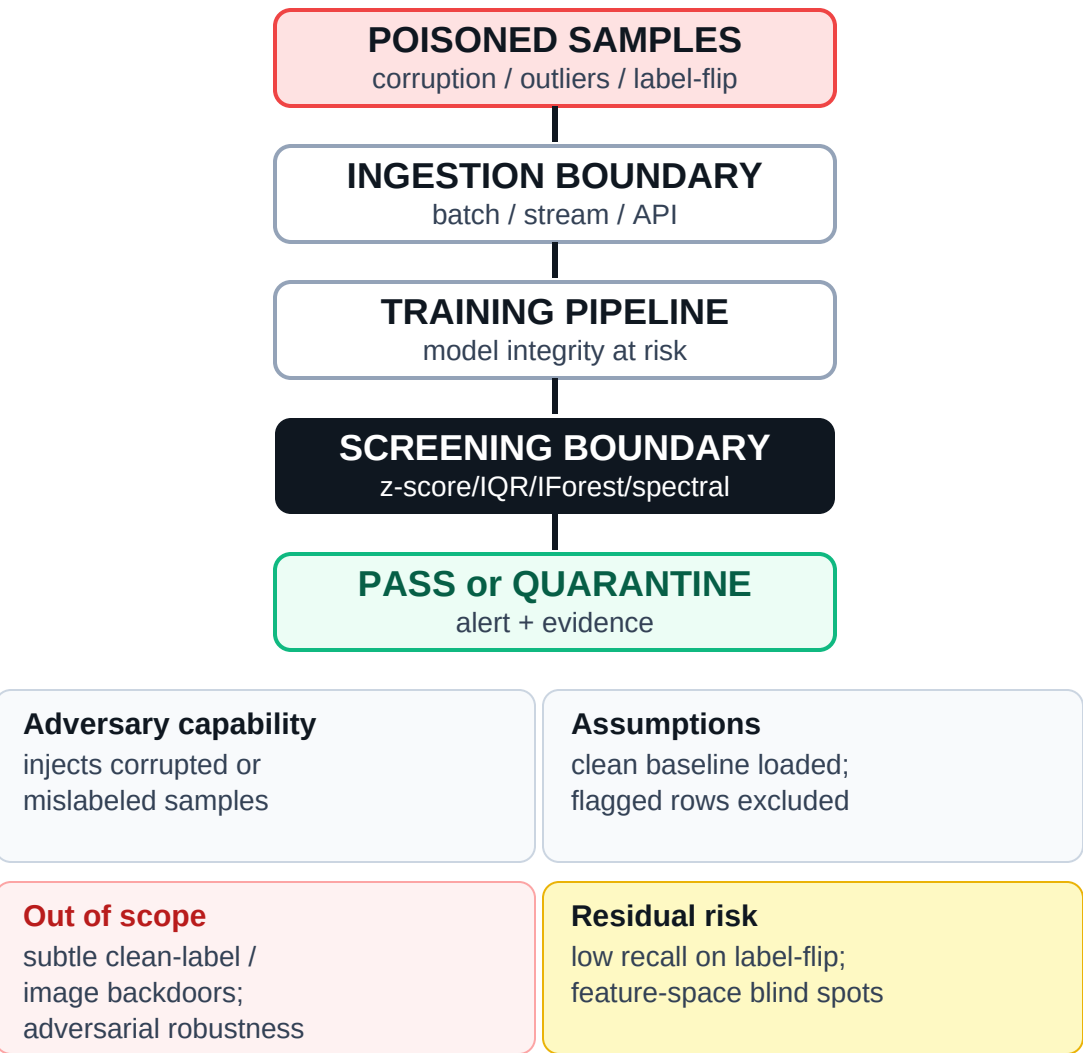
RESEARCH QUESTION

How effectively can statistical screening identify poisoned or corrupted training samples before they enter the training pipeline?

01 • THE FAILURE MODE

Poisoned or corrupted samples enter training silently. A vendor feed drifts or a shared table gets bad rows, and precision quietly drops over retraining cycles — the pipeline never errors, so nobody checks the data. A screening layer at ingestion (MITRE ATLAS AML.T0020) can catch gross corruption — but not subtle attacks.

02 • THREAT MODEL



03 • OBJECTIVE

Determine how well an unsupervised statistical ensemble screens ingestion data — and measure explicitly where it fails.

- 01 – Z-score + IQR + IsolationForest
- 02 – Spectral signatures (label-aware)
- 03 – Majority-vote ensemble + quarantine
- 04 – Streaming (Welford) + drift

HERO • INGESTION SCREENING ENSEMBLE

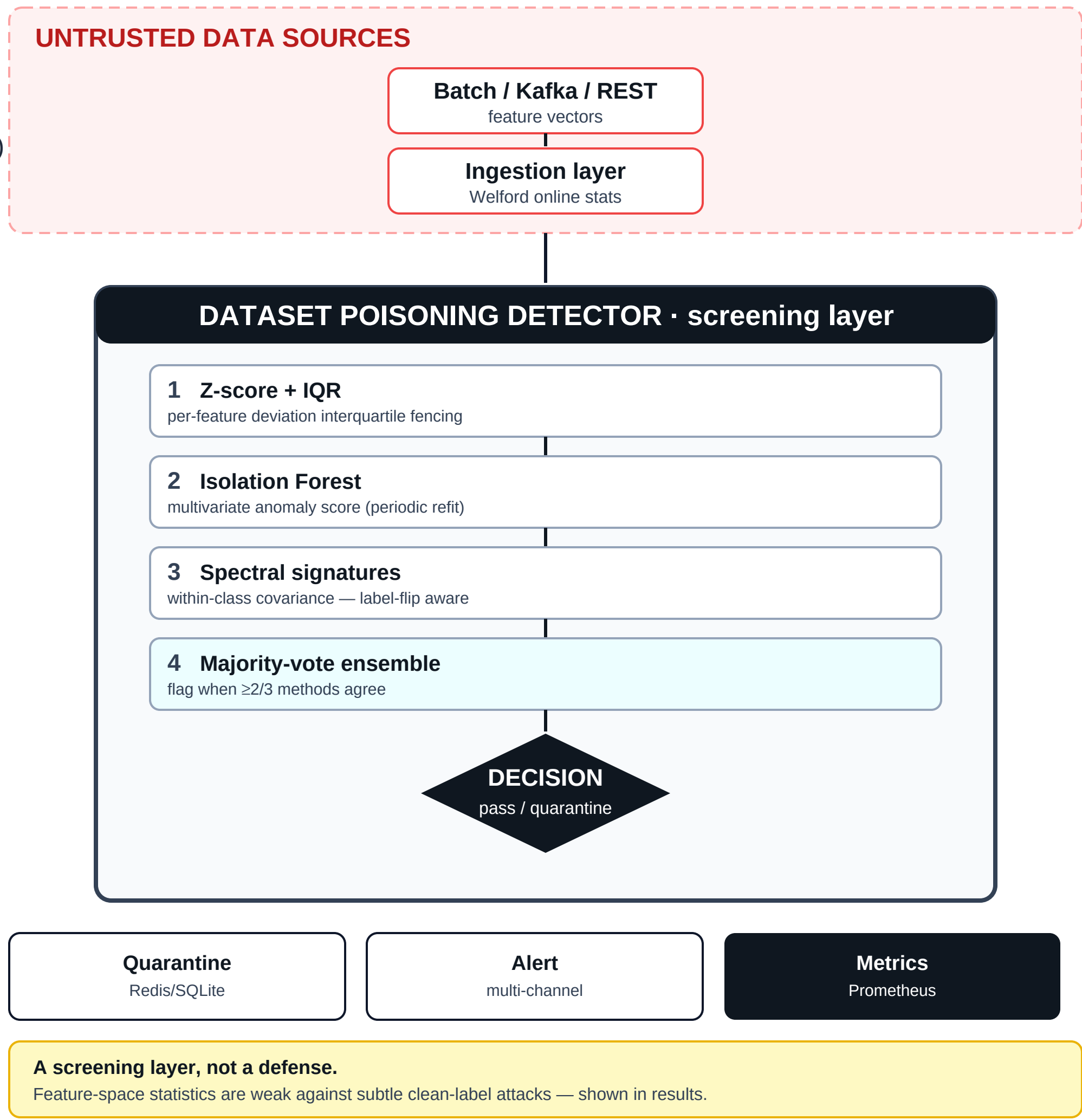
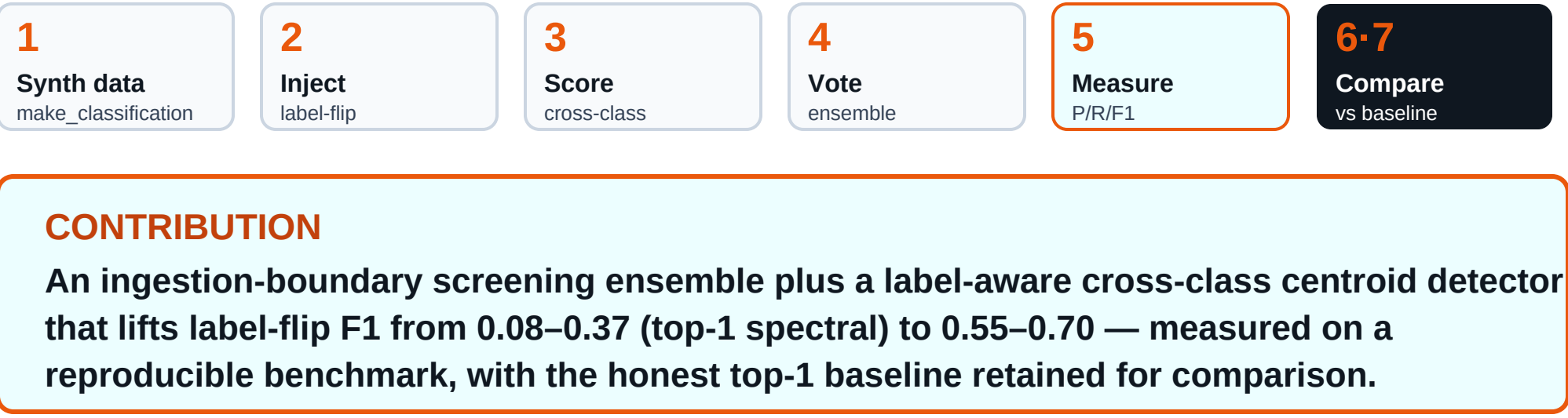


Figure 1. Ensemble screening at the ingestion boundary; spectral analysis targets label-flip attacks feature-space methods miss.

METHOD • EFFICACY BENCHMARK



RESULTS • LABEL-FLIP DETECTION (measured)

0.70
BEST F1 (cross-class)

0.55
F1 @5% poison

Label-flip benchmark, seed 2018, 2000×100 synthetic. Cross-class centroid detector; top-1 baseline kept for honesty.

Cross-class F1 vs contamination	
@ 5% poison	0.55
@ 10% poison	0.60
@ 20% poison	0.70

F1 BY METHOD (@20% contamination)	
Cross-class F1 (x100)	70%
Top-1 baseline (x100)	37%
Ensemble (x100)	23%

Cross-class centroid beats top-1 spectral and the feature-space ensemble on 3/3 rates. spectral

BENCHMARK PROVENANCE
results/spectral_benchmark.json (committed).
Tran et al. 2018 setup. Top-1 baseline (0.08/0.23/0.37) retained in the JSON for transparent comparison.

NOT ESTABLISHED BY THIS REPO
High recall on label-flip attacks. Detection of subtle clean-label / image backdoors. Robustness guarantee.

SECURITY BOUNDARY

PROTECTS / DETECTS

- gross outliers
- obvious corruption
- distribution shift
- replay (fingerprint)
- provenance logging

DOES NOT ESTABLISH

- subtle clean-label attacks
- image backdoors
- targeted poisoning
- robustness proof

SCREENING ≠ DEFENSE

LIMITATIONS

- 1. Cross-class F1 0.55–0.70; still misses ~30–45% of flips.
- 2. Feature-space methods blind to clean-label attacks.
- 3. Streaming stats are per-feature; correlations lost.
- 4. Synthetic benchmark ≠ real pipeline data.
- 5. First-pass filter, not adversarial robustness.

FUTURE WORK

- Evaluate on representative real pipeline data.
- Learned label-aware detectors.
- Multivariate streaming statistics.
- Clean-label attack detection research.
- Calibrated thresholds per dataset.

CONCLUSION

The tool screens ingestion data with a 4-method ensemble and a label-aware cross-class detector that lifts label-flip F1 from 0.08–0.37 (top-1) to 0.55–0.70, shown beside the honest baseline. It is a strong first-pass filter, not an adversarial-robustness proof.

REPRODUCE
pytest tests/
python benchmark/cifar10_label_flip_benchmark.py